

Script-Controlled ACL – Restrict Record Access Based on Field Value

2. Requirement Analysis – Completed Answer

2.1 Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User and role setup	Create EEE User; create roles bb1, bb2, bb3, bb4; assign roles to EEE User.
FR-2	Institution Details data model	Create u_institution_details with Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description.
FR-3	Record-level READ security	Permit bb1 users to read EEE-branch records; deny users without the required role; permit admin full access.
FR-4	Record lifecycle security	CREATE requires bb2; WRITE requires bb3; DELETE requires bb4.

2.2 Non-Functional Requirements

NFR	Requirement	Description
NFR-1	Usability	Authorized users can open the Institution Details list and work according to their permissions.
NFR-2	Security	ACLs, required roles, Branch condition and script enforce access.
NFR-3	Reliability	The same ACL rules should consistently apply during normal access.
NFR-4	Performance	Normal ACL evaluation should complete during record access; no numeric target is supplied.
NFR-5	Availability	Depends on ServiceNow instance availability; no target is supplied.
NFR-6	Scalability	The ACL pattern can be extended; no quantitative target is supplied.

2.3 User Stories

User Type	User Story	Acceptance Criteria	Priority
EEE User / bb1	As an authorized EEE user, I can view EEE branch records.	EEE records visible; other branch records not visible through the READ rule.	High
User without bb1	As an unauthorized user, I cannot view Institution Details records.	Records are not viewable through the configured READ ACL.	High
Administrator	As an administrator, I can view all Institution Details records.	All branches are visible.	High
bb1 + bb2	As a user with bb1 and bb2, I can create records.	New operation is available according to CREATE ACL.	High
bb1 + bb2 + bb3	As a user with bb1, bb2 and bb3, I can edit EEE records.	EEE records can be edited according to WRITE ACL.	High
bb1 + bb2 + bb3 + bb4	As a user with all four roles, I can delete permitted records.	Delete operation is available according to DELETE ACL.	High

2.4 Data Flow – textual DFD

User → ServiceNow role/authentication context → ACL evaluation → u_institution_details → permitted records returned. READ evaluates the admin/bb1 script and the Branch = EEE condition; CREATE/WRITE/DELETE use their respective role ACLs.